Terminal 2: Run Publisher

python3 publisher.py

8. Result

Subscriber Terminal:

Connected
Received: ON
LED ON
Received: OFF
LED OFF

Publisher Terminal:

text
Enter ON/OFF : ON
Published : ON
Enter ON/OFF : OFF
Published : OFF

PRACTICAL – 14

1. Aim

Design a web dashboard using Flask. Control relay module via web.

2. Prior Concepts

i) Concept of Python Programming.

ii) Working of relay modules.

iii) Basic networking concepts.

3. New Concepts

i) IoT-based remote appliance control.

ii) Interfacing relay modules with Raspberry Pi.

iii) Web-based appliance automation.

iv) Flask web server programming.

v) GPIO control using Python.

vi) Smart home automation concepts.

4. Hardware Required

i) Relay Module (2-channel)

Input IMAGE

ii) Bulb

iii) DC Fan

iv) Jumper Wires

5. Procedure

Pin Connection

Input IMAGE

i) Install Python3 on Raspi.

Write the below code in command prompt:

sudo apt install idle3

pip install flask

pip install RPi.GPIO

6. Implementation with Python Code

Code:

```
from flask import Flask

import RPi.GPIO as GPIO

app = Flask(__name__)

# GPIO Setup
GPIO.setmode(GPIO.BCM)

LIGHT = 17

FAN = 27

GPIO.setup(LIGHT, GPIO.OUT)

GPIO.setup(FAN, GPIO.OUT)

GPIO.output(LIGHT, GPIO.LOW)

GPIO.output(FAN, GPIO.LOW)



@app.route('/')
def home():
    return """
    <h1>IoT Home Automation</h1>
```

```
    <h3>Light</h3>

    <a href="/light/on">ON</a><br><br>

    <a href="/light/off">OFF</a><br><br>

    <h3>Fan</h3>

    <a href="/fan/on">ON</a><br><br>

    <a href="/fan/off">OFF</a>
    """


@app.route('/light/on')
def light_on():
    GPIO.output(LIGHT, GPIO.HIGH)
    return "Light ON"


@app.route('/light/off')
def light_off():
    GPIO.output(LIGHT, GPIO.LOW)
    return "Light OFF"


@app.route('/fan/on')
def fan_on():
    GPIO.output(FAN, GPIO.HIGH)
    return "Fan ON"


@app.route('/fan/off')
def fan_off():
```

```
    GPIO.output(FAN, GPIO.LOW)
    return "Fan OFF"


if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)
```

7. Execute the Above Program

i) Run the above Program.

ii) Find Raspi IP Address, by executing below command on command prompt:

hostname -I

Eg: 192.168.1.10

iii) Open Web Interface on mobile or laptop and enter the IP address in browser.

Eg: http://192.168.1.10:5000

8. Result

The web dashboard using Flask is designed successfully to control the relay module via the web.

PRACTICAL – 15

1. Aim

Create two factor authentication system by interfacing fingerprint sensor, RFID reader with Raspberry Pi.

2. Prior Concepts

i) Concept of Python Programming.